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A predictive analytics approach for forecasting bike rental demand

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ABSTRACT

The demand for rental bikes in urban areas fluctuates, leading to localized surpluses and shortages. To address this challenge, effective bike relocation strategies are essential for ensuring equitable distribution and maximizing customer satisfaction. This study aims to employ advanced machine learning techniques to forecast bike rental demand in urban areas, thereby enhancing the efficiency and accessibility of bike rental services and contributing to sustainable urban mobility. The study comprehensively analyzes various influencing factors using machine learning models, including Ordinary Least Squares regression, MLP Regression, Gradient Boosting Regression, Random Forest Regression, Polynomial Regression, and Decision Tree Regression. The primary objective is to identify the most accurate predictor by comparing key metrics such as *R*-squared (R2), Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Pearson Correlation Coefficient. Insights gained from this analysis aid in identifying influential variables and ensure the development of resource-efficient and adaptable models, leading to more informed decision-making for rental bike businesses. Additionally, future research directions involve the implementation of artificial intelligence technology to predict overall bike demand based on urban cities' criteria, including the number of national and international tourists. By addressing these objectives, this study seeks to provide valuable insights and tools for rental bike businesses to optimize operations, make strategic decisions, and enhance customer experience in competitive urban markets.

1. Introduction

The evolution of modern urban transportation, driven by burgeoning urban populations, has been substantial. This paper's data set is based on the urban city of Seoul, in South Korea. Seoul, with a population of nearly 9.86 million in 2015, grappled with the challenges of urbanization. In response, the Korean government introduced a rental bike system in 2015. This initiative aimed to alleviate traffic congestion, mitigate air pollution, and reduce high energy costs while fostering a healthier society and elevating citizens' quality of life. This innovative bike rental system, Seoul Public Bike or Tdareungi, is a pivotal advancement in the ever-changing transportation paradigm. Mobile applications have been instrumental in this transformation, facilitating real-time bike tracking, secure locking and unlocking, seamless payments, route guidance, maintenance reporting, and user engagement. These applications empower users with convenience and data-driven insights, ultimately enhancing the efficiency and accessibility of modern urban transportation. From the User Base vs. Commuters perspective, the 96.9% user satisfaction rate is related to those who have already adopted the bike-sharing system. It reflects how satisfied these users are with the service. However, not all residents are bike users. The 36.3% figure indicates the portion of the entire population who have integrated it into their daily commuting routine. This survey can serve as an encouragement for residents who have not yet adopted the system. When they see that a significant portion of their fellow residents are incorporating it into their daily commutes, it might motivate more people to try it. Furthermore, it underscores the diversity of the bike-sharing system's use cases, with an additional 26.8% of residents using the service for recreational purposes. Understanding the percentage of daily commuters using the system is invaluable for urban planners as it sheds light on the potential for reducing traffic congestion and public transport strain [1].

The demand for rental bikes has seen a significant increase over the years, with daily rentals reaching 87,800 in 2021, up from 65,000 the previous year. However, challenges persist due to fluctuating demand across various periods and regions. These discrepancies occasionally lead to shortages or surpluses of available bikes, affecting both the revenue of service providers and the convenience of users. Tourists in Seoul often opt to rent bikes to explore the city. Bike rental is an affordable and accessible option for everyone. Seoul is increasingly embracing bikers, boasting over 240 kilometers of riverside bike paths extending throughout the city and nearby towns. While dedicated bike lanes within the city may be limited, there are smaller side streets

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where biking is much more convenient and enjoyable [2]. The best time to visit South Korea is typically from spring to late June, with pleasant weather. Spring, particularly in April, is famous for cherry blossoms in Seoul. South Korea is a popular tourist destination, with 11.1 million international visitors in 2012, mainly from nearby East Asian countries. Seoul is a key tourist hub. Domestic tourism is growing, with 238.3 million locals traveling in 2015, a 4.9% increase from the previous year, and spending ₩14.4 trillion in 2014. Korean residents' interest in overseas travel has been steadily rising, with 16.1 million outbound travelers and $19,469.9 million in expenditure in 2014 [3]. This paper aims to illuminate the reasons behind demand fluctuations in bike-sharing start-ups through a comprehensive analysis of datasets from Seoul's bicycle-share program and surveys. By examining correlations between independent variables such as temperature, time of day, season, and holidays and dependent variables like rented bike counts, the study leverages machine learning techniques, including OLS regression, random forest regression, and Gradient Boosting regression, to predict demand patterns. The insights garnered from this analysis provide a foundation for informed decision-making. By strategically allocating bicycles based on geographical nuances, the bike-sharing system can potentially address both shortages and surpluses, resulting in a more optimized and user-friendly experience [4–7].

2. Literature review

In today's urban environments, cities are evolving rapidly, especially regarding transportation. Local governments are actively working on plans to ease traffic jams, reduce air pollution, and cut down on energy usage. They aim to create healthier communities and improve residents' overall quality of life. One standout solution gaining popularity is bike-sharing systems. These systems offer a convenient way to get around and encourage healthier lifestyles, making them a win–win for urban dwellers and the environment. Various machine learning algorithms have been used to predict the rental bike demands.

Davis, Kutschinski, and Yang (2022) conducted a study on predicting hourly bike demand in Seoul's bike-sharing program. They found the LSTM Lag 12 model to be the most accurate with a Mean Absolute Error (MAE) of 78 bikes per hour. The study recommends immediate implementation of this model for short-term demand forecasting but acknowledges limitations related to spatial variation in demand [8]. Arora, Nandasana, Mali, and Wani's (2022) study focuses on Seoul's bike-sharing program, utilizing data from the Seoul Bike and Capital Bikeshare programs. They emphasize a rule-based regression predictive model, prioritizing environmental benefits and reducing waiting times [9]. Sathishkumar V E and Yongyun Cho's work adopts a rule-based regression predictive model approach, emphasizing specific statistical models like CUBIST and Regularized Random Forest for bike-sharing demand prediction. Their focus lies on presenting results and insights from implementing these models, primarily analyzing specific statistical models' performance in explaining rental demand variance [10].

Wenzhen Jia et al. (2019) proposed a hierarchical prediction model for shared bicycle transportation, addressing station imbalance issues. The model employs clustering, gradient-boosting regression, and multi-similarity-based inference. Validation on NYC and D.C. bike-sharing systems showed superior performance, reducing check-out and check-in prediction errors by 8%–22% and 2%–3%, respectively, compared to other methods [11]. Xinwu Qian and collaborators explore enhancing urban on-demand mobility services (OMS) like taxis and ride-sharing platforms. They introduce a novel boosting Gaussian conditional random field (boosting-GCRF) model to accurately predict short-term future demand using historical data. Through extensive experiments, they demonstrate the model's superiority over existing algorithms, achieving a mean absolute percentage error of just 14%. The boosting-GCRF model proves resilient to demand anomalies and effectively captures the distribution of short-term taxi demand. This research offers valuable insights for improving the efficiency of urban transportation services [12].

Zhou et al. from Southwest Jiaotong University introduce a Markov Chain-based demand prediction model for bike-sharing systems' stations. This model aims to accurately forecast transfer demand, crucial for optimizing station layouts and bike allocation. Traditional traffic demand prediction methods are inadequate for bike station forecasting due to unique station characteristics. The proposed model is evaluated using field data from Zhongshan City's bike-sharing system, demonstrating superior accuracy and generalization compared to existing methods [13].

This study investigates various machine learning models, encompassing the Hybrid Autoencoder-Regressor Model, Multi-Layer Perceptron Regressor, Gradient Booster Regression, Random Forest Regression, Transformer Model, and Deeper Neural Network Model. These models undergo evaluation using performance metrics such as *R*-squared, Mean Squared Error, Mean Absolute Error, Root Mean Squared Error, and Pearson Correlation Coefficient. Findings indicate that Random Forest Regression consistently outperforms other models across multiple metrics, attributed to its capacity to apprehend intricate non-linear relationships within the data. The study underscores the importance of optimizing resources and enhancing operational efficiency in bike rental services, proposing utilizing machine learning models like Random Forest Regression to enhance predictive accuracy notably. Future research endeavors to investigate additional machine-learning models and external factors influencing bike rentals, with the objective of further refining predictive accuracy. Furthermore, insights gleaned from prior studies on bike demand analysis and prediction contribute to advancements in forecasting precision and operational effectiveness within bike-sharing systems [14,15].

3. Process of rental bike demand

The Rental Bike Demand Analytics outlines a structured progression. It begins with 'Requirements Gathering,' wherein stakeholders meticulously define project objectives, scope, and inquiries through engaged interactions. Following this, the 'Analytics Approach' identifies methodologies that encompass the judicious selection of relevant techniques to address identified analytical challenges effectively. Upon finalizing the approach, the project seamlessly transitions to the 'Data Collection' phase, where meticulous data acquisition occurs. The dataset, sourced from Kaggle, serves as the cornerstone, ensuring a dependable foundation for analysis. Rigorous preprocessing ensues, encompassing the elimination of duplicates, managing null values, and aligning formats for impeccable data quality. This meticulous approach safeguards the integrity of the dataset. With refined data in hand, the 'Data Exploration' stage emerges, utilizing visualization, statistical analysis, and exploratory techniques to unveil patterns, relationships, and insights. This leads to the 'Modeling' phase, which initiates the creation of descriptive models and three prescriptive models using fitting algorithms. Data is divided into 80% for training and 20% for testing sets, with models refined, optimized, and evaluated against predetermined objectives. This is a standard practice in machine learning to assess the model's performance on unseen data and avoid overfitting. These machine-learning algorithms are proficient in creating, training, and evaluating the Hybrid Autoencoder-Regressor Model, Multi-Layer Perceptron Regressor, Gradient Booster Regression, Random Forest Regression, Transformer Model, and Deeper Neural Network Model. These models calculate key evaluation metrics such as *R*-squared (R2), Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Pearson Correlation Coefficient, which are essential for assessing the model's accuracy and performance [16,17].

The process progresses through different stages: "Evaluation" assesses how well the model performs by comparing it against industry standards. If it meets the criteria, it proceeds to "Deployment" where it is applied in real-world scenarios. Collaboration among experts and

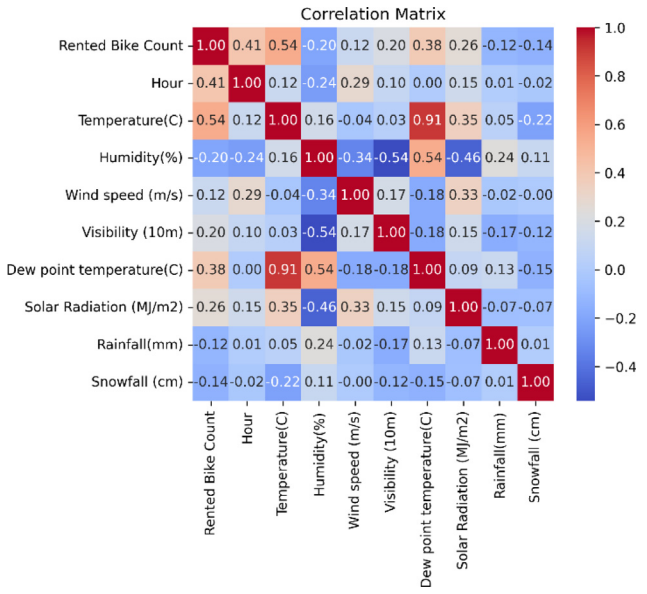


Fig. 1. Correlation of rented bike variables.

data professionals facilitates incremental improvements. Ensuring accurate data from the outset enhances predictive accuracy, as seen in a project predicting city bike demand. Various Python tools were employed for data analysis and prediction, such as Pandas for data organization, NumPy for mathematical operations, Seaborn for pattern visualization and stats models for statistical comprehension. Ordinary Least Squares (OLS) regression was initially attempted, but more sophisticated methods like Gradient Boosting Regressor and Random Forest Regressor were pursued for better outcomes. Model performance was evaluated using metrics like r2_score and mean absolute error, while Pearson from SciPy Stats was utilized to grasp variable correlations. The data underwent thorough analysis through meticulous tool usage, leading to the development of effective models and insightful predictions. This study aims to give rental bike businesses valuable insights and tools to optimize operations, make strategic decisions, and enhance customer experience in competitive urban markets. It achieves this by comparing various machine learning models and justifying the selection of the most suitable model for accurate business forecasting decision-making [18–20].

4. Correlation test

The correlation analysis reveals exciting relationships among the variables in the dataset. The correlation coefficient, which ranges from −1 to 1, provides insights into the strength and direction of these relationships. A positive coefficient denotes a positive linear relationship, while a negative coefficient indicates a negative linear relationship. A coefficient value of 0 signifies the absence of a linear relationship between the variables [16]. Upon examining the correlation of the bike rental data set variables, Fig. 1 shows that most variables display either low or negligible correlations. However, two pairs of variables stand out. Firstly, the temperature and dew point temperature exhibit a substantial positive correlation coefficient of 0.91. This high correlation suggests that the dew point temperature also tends to rise as the temperature increases. Secondly, the correlation between humidity and dew point temperature is 0.54, indicating a moderate positive connection between these variables. Conversely, wind speed, visibility, solar radiation, rainfall, and snowfall demonstrate weak or insignificant correlations with the number of rented bikes [21–23].

Exploring strong correlations uncovered through correlation analysis offers valuable insights into rental bike demand analysis. By scrutinizing variables with significant positive or negative correlations, analysts can develop a deeper comprehension of the factors influencing bike rental demand. For instance, within the realm of weather variables, robust positive correlations between temperature and bike count may signify that warmer weather correlates with increased bike rentals. Conversely, robust negative correlations, such as those between rainfall and bike count, might imply that rainy weather deters bike usage. Understanding these correlations enables more precise predictive modeling and informed decision-making to optimize bike rental services. Furthermore, delving into strong correlations can reveal potential areas for intervention or enhancement, such as adjusting rental pricing or refining marketing strategies based on weather forecasts. In essence, leveraging robust relationships unveiled through exploratory analysis enriches the efficacy of rental bike demand analysis, fostering data-driven strategies to cater to customer needs and maximize operational efficiency.

5. Data insights through exploratory data analysis

This section introduces an analytics suite designed to address the demand for rental bikes, illustrating the workflow for tackling the issue. The methodology comprises essential stages: requirement gathering, data collection, data exploration, data preprocessing, modeling, and model evaluation. The primary aim of the analytics suite is to facilitate Exploratory Data Analysis (EDA) and prediction. EDA plays a crucial role by delving into the acquired dataset to gain initial insights and a comprehensive understanding of the data. It employs visualization and summary techniques to reveal patterns, trends, relationships, and potential outliers. EDA is instrumental in predicting rental bike counts as it aids in data understanding, identifying correlations between predictor variables and target variables, detecting outliers, performing feature engineering, and conducting data preprocessing. Additionally, it assists in model selection, evaluation, and data visualization, providing a comprehensive approach to predicting rental bike demand.

Predictive analytics, a vital component of the analytics suite, focuses on examining rental bike demand based on various factors, including temperature, time of day, season, holiday occurrences, functional days, rainfall, snowfall, humidity, wind speed, visibility, dew point temperature, and solar radiation. By incorporating predictive analytics, the suite empowers stakeholders to gain valuable insights into bike rental trends and make informed decisions to optimize rental bike services. Gathering information is crucial for conducting predictive analytics, including data about the urban city, its rental bike system, population statistics, tourist arrival figures, optimal tourism seasons, seasonal temperature variations, humidity levels, visibility conditions, wind speed patterns, and solar radiation levels, as well as the distinct seasons characterized by rainfall and snowfall. Integrating these diverse datasets provides a robust foundation for predictive modeling, enabling a comprehensive assessment of the factors influencing rental bike demand in the urban city [14,15,24–26].

5.1. Distribution of temperature for rented bike counts

Distributions are a fundamental aspect of Exploratory Data Analysis (EDA). They provide a quick overview of data characteristics, reveal patterns and anomalies, and guide further analysis and modeling decisions. In the context of the urban city's bike rental demand analytics, distributions help understand how the demand for rental bikes is influenced by various factors, such as temperature. The urban city's climate, with its distinct seasons and varying temperatures, plays a significant role in driving bike rental demand. Spring and autumn, characterized by mild and comfortable weather, create ideal conditions for bike riding. These delightful climates encourage residents and tourists to explore the city on bikes, resulting in heightened demand for rental bicycles during these periods.

Analyzing the distribution of temperature and its correlation with rented bike counts is particularly insightful. Fig. 3 illustrates the frequency of bike rentals across different temperature ranges, pinpointing

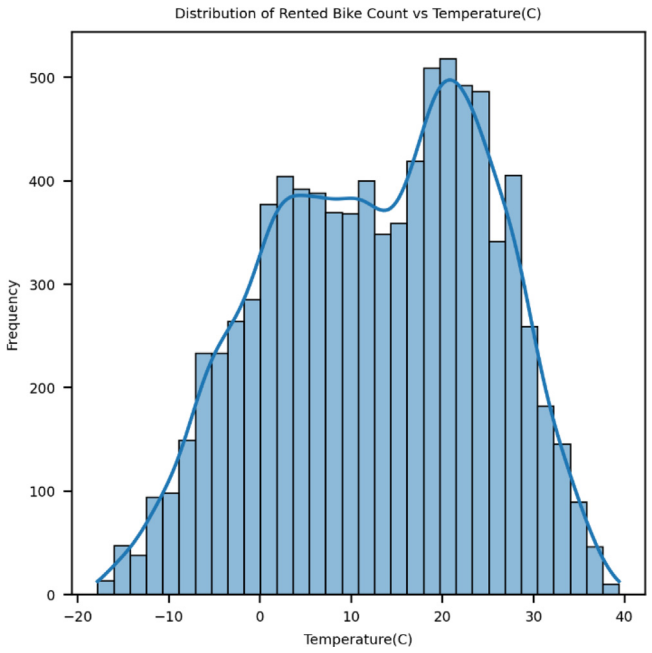


Fig. 2. Distribution of rented bike count vs temperature.

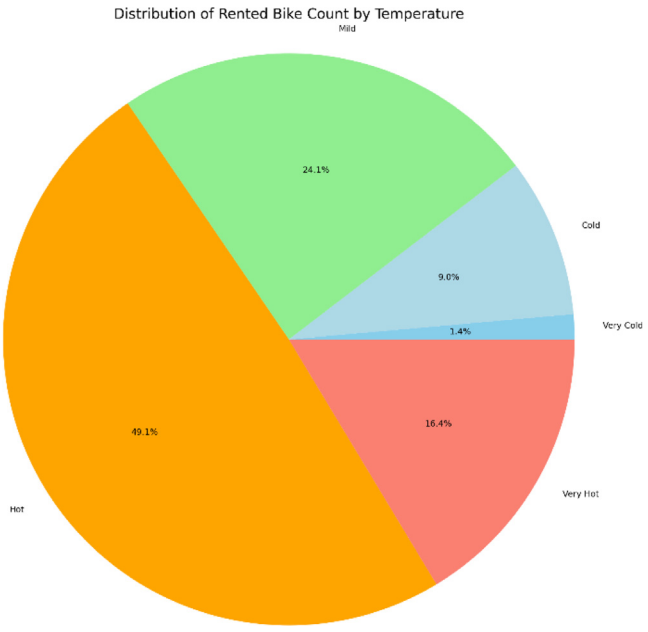


Fig. 3. Total rented bike count vs temperature.

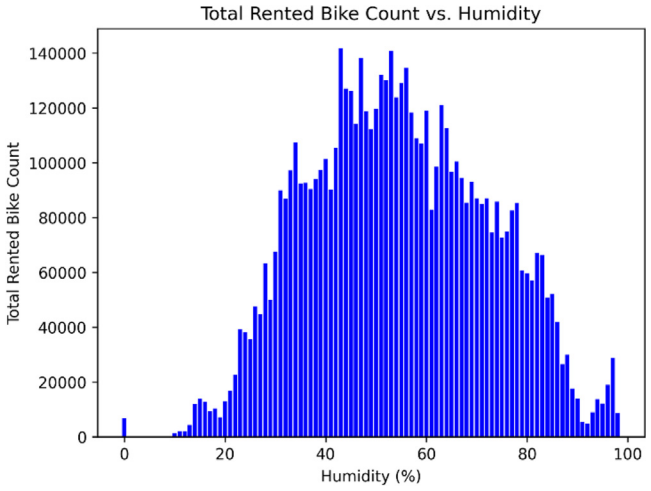


Fig. 4. Total rented bike count vs humidity.

intervals with increased rental activity. This data aids in recognizing temperature-related trends in rental demand peaks and contributes to optimizing bike-sharing services. Additionally, Fig. 2 underscores a strong correlation between temperature and bike rental demand, with peak demand occurring within the 20–25º C temperature range.

The city experiences a range of temperature conditions throughout the year. The winter months of December to February are very cold, with temperatures between −5 °C and 4 °C. November and March fall into the cold category, with temperatures ranging from 1 °C to 11 °C. In contrast, April and October offer mild weather, with temperatures between 13 °C and 19 °C. May and September are considered hot months, with temperatures from 18 °C to 25 °C. The summer months of June, July, and August are very hot, with temperatures ranging from 22 °C to 29 °C. These temperature ranges define the climate conditions in the corresponding urban city, impacting various aspects of daily life, including preferences for outdoor activities like bike riding. Even during warmer weather, Seoul records significant bike rentals, while colder months witness a decline. Seoul's climate not only influences bike rental demand but also fosters a culture of healthy living and outdoor exploration, thanks to favorable conditions for outdoor activities and well-being in the city's warm weather.

5.2. Humidity's impact on rental bike preferences

In the examined dataset of Seoul's bike rental system, Fig. 4, titled "Total Rented Bike Count vs Humidity", draws attention to the influence of humidity levels on bike rental patterns. Notably, it reveals a striking trend in bike rentals corresponding to varying humidity ranges. The data indicates that the highest number of bike rentals occurs when humidity falls within the range of 40%–60%. During these specific periods, a substantial cumulative bike rental count of 141,773 was recorded, signifying a heightened interest in and reliance on the city's bike-sharing service. This figure represents about 2.30% of the total bikes available for rental in Seoul. This finding underscores the significance of humidity as a weather-related factor in shaping people's preferences for utilizing rental bikes. Higher humidity levels may encourage individuals to choose indoor activities or alternative modes of transportation over biking, resulting in lower rental demand. In contrast, when humidity is within the 40%–60% range, it aligns with favorable biking conditions, resulting in increased demand for rental bikes. These insights highlight the complex interplay between humidity levels, weather conditions, and individuals' choices regarding bike rentals in Seoul.

5.3. Distribution of wind speed for rental bike count

Cycling can be greatly influenced by wind conditions, with headwinds posing resistance and tailwinds providing acceleration. Headwinds can be particularly challenging, demanding increased effort from riders, especially when cycling uphill or on open, wind-exposed roads. These headwinds can significantly reduce cycling speed, leading to longer travel times. On the contrary, the presence of tailwinds offers a welcome boost, propelling cyclists forward and improving overall efficiency. This analysis of Seoul's bike rental system delves into the relationship between wind speed and individuals' choices when it comes to renting bikes. Interestingly, the data reveals a preference for cycling during mild wind conditions, with approximately 295,750 people, which accounts for 4.79% of the total rentals, opting for wind

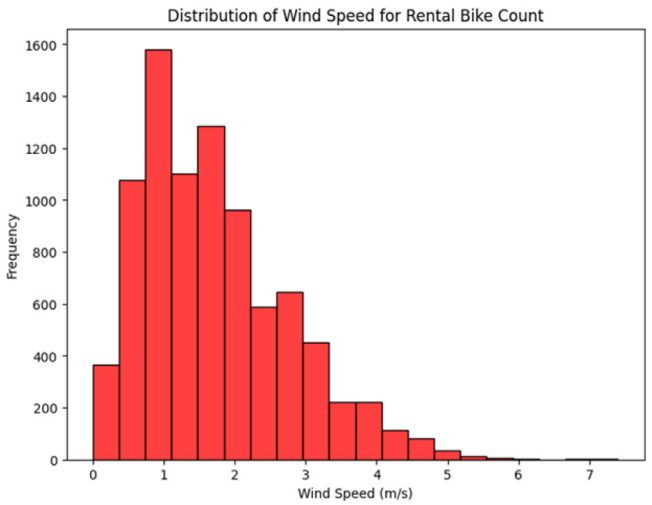


Fig. 5. Distribution of rented bike count vs wind speed (m/s).

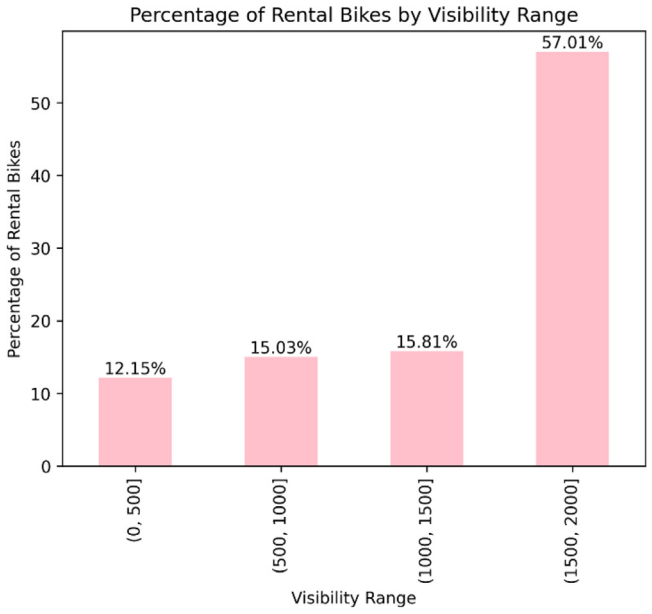


Fig. 6. Total rented bike count vs visibility.

speeds between 1 to 2 meters per second. This finding highlights the appeal of biking in gentle winds, offering valuable insights for bike-sharing services in optimizing cycling experiences. It underscores the notion that moderate wind conditions are conducive to promoting biking as a health-conscious and enjoyable mode of transportation (see Fig. 5).

5.4. Visibility's impact on bike rental counts

Visibility is a critical factor that significantly influences bike rental patterns, affecting both safety and overall riding experience. Adequate visibility enables cyclists to navigate roads and surroundings more effectively, reducing the risk of accidents and ensuring a more enjoyable biking experience. Conversely, low visibility can make cycling seem riskier, leading to reduced bike rentals. Therefore, raising awareness about the importance of visibility and its positive impact on bike rentals is essential, especially during favorable weather conditions. Fig. 6 reveals that the lowest visibility range (0–50 m) accounted for 6.15% of rented bikes between 2017 and 2018. In contrast, the highest

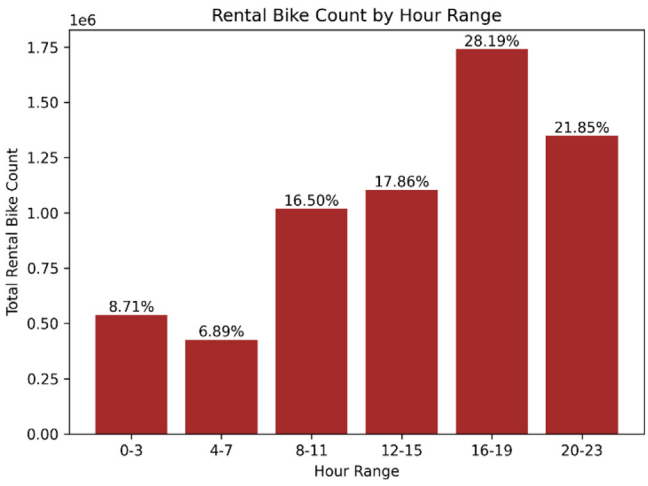


Fig. 7. Total rented bike count vs hour.

visibility range (150–200 m) witnessed a substantial rental count of 63.54% during the same period, spanning all four seasons, including summer, autumn, spring, and winter. Notably, during winter, factors such as low temperatures, mist, fog, and snowfall can contribute to reduced visibility conditions. This figure represents 63.54% of the total rented bikes, highlighting the strong relationship between visibility and bike rentals.

5.5. Distribution of hours for rental bike count

Seoul, the bustling capital of South Korea, stays active day and night. Time is crucial for businesses, including bike rentals. The self-access bike rental system operates 24/7, fitting the city's rhythm. People can use it any time they need, aligning well with Seoul's nonstop energy. Insights3 provides insights into distinct usage patterns within the dataset of bike rentals, segmented by time of day. During the late night and early morning hours (Hour Range 0–3), approximately 8.71% of total bike rentals occur, indicating some level of demand during these quieter periods. As the morning progresses (Hour Range 4–7), the percentage of rentals decreases slightly to 6.89%, suggesting a gradual increase in activity as the city awakens. By mid-morning (Hour Range 8–11), the percentage rises to 16.50%, indicating a growing preference for rental bikes as people begin their daily routines. Throughout the afternoon (Hour Range 12–15), the percentage remains steady at 17.86%, reflecting sustained usage for commuting or leisure. The evening rush hour (Hour Range 16–19) witnesses a significant spike, with rentals comprising 28.19% of the total, underscoring the popularity of bikes as a mode of transport during busy commute times. Even during the late evening hours (Hour Range 20–23), rental bike usage remains substantial at 21.85%, highlighting continued demand for urban mobility options throughout the day.

5.6. Influence of rainfall vs rental bike count

The analysis of bike rentals about rainfall uncovers intriguing trends. Fig. 10 illustrates that rented bike numbers vary on the top 15 rainfall measures during rainy periods, reaching a low of 10 with 15.5 mm and a high of 596 with 21.5 mm. This dynamic indicates a decreased preference for biking when it rains, suggesting that people avoid biking during inclement weather due to potential discomfort and safety concerns. Alternative transportation options are available on rainy days. This insight holds significance for bike-sharing companies and urban planners alike. Understanding how people's behavior shifts with weather conditions enables tailoring services to meet their needs.

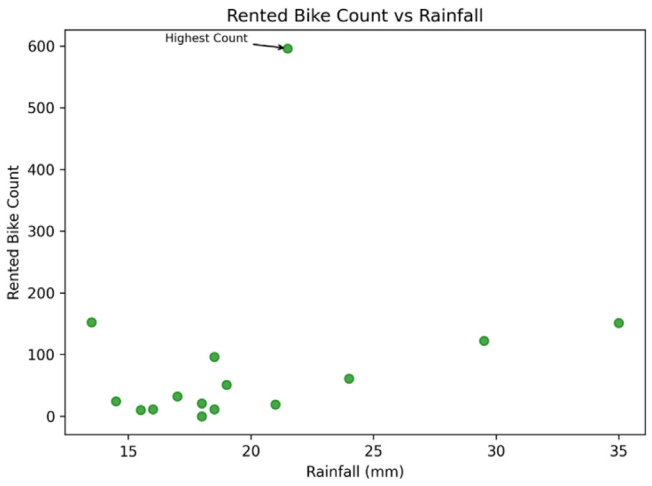


Fig. 8. Influence of rainfall vs rented bike count.

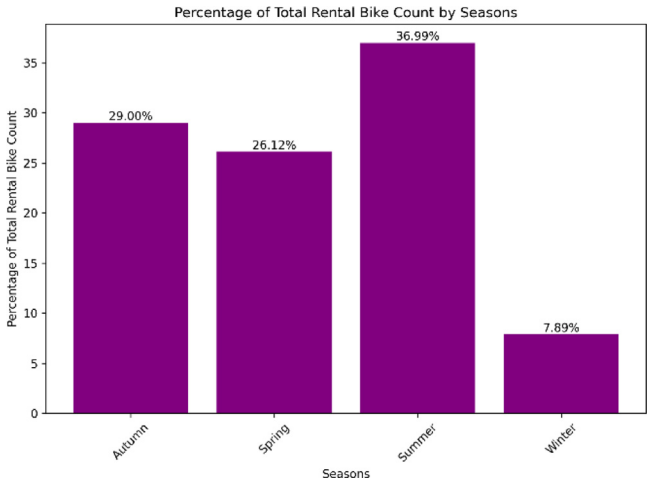


Fig. 9. Percentage of rented bikes vs seasons.

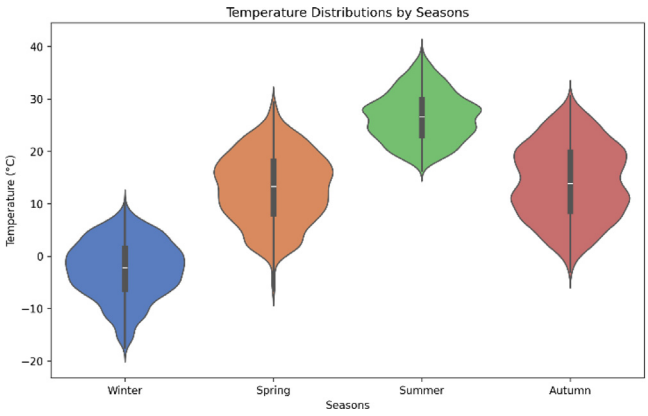


Fig. 10. Temperature distributions by seasons.

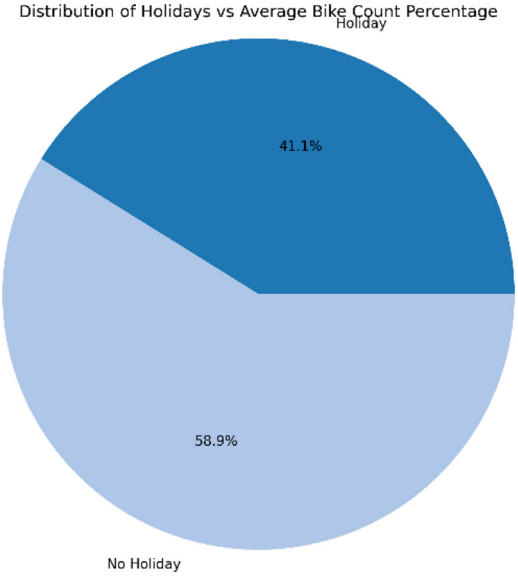


Fig. 11. Distribution of average bikes rented on holidays vs non-holidays.

Furthermore, rain's impact extends beyond usage; it affects the condition of bike components, with prolonged exposure leading to wear and tear. Consequently, maintenance costs may rise as bike rental companies strive to upkeep their fleet during the rainy season. The influence of rainfall on bike rentals in Seoul tends to be unfavorable, causing a decrease in rental counts and introducing operational complexities for bike-sharing enterprises (see Figs. 7 and 8).

5.7. Seoul seasons vs rental bike count

The column plot shows percentage rental bike counts in Seoul exhibit variations across different seasons [18]. During the autumn, the percentage bike count is 29%, suggesting that biking remains a popular choice among residents and visitors as the weather transitions from warm to cool. In the spring, the average percentage is 26.10%, reflecting a slightly lower preference for biking as the city embraces the blooming season. As summer arrives, the average rental bike count percentage sees a noticeable increase to 36.99%, indicating a surge in biking activity as people take advantage of the pleasant weather for outdoor exploration. In contrast, the winter season witnesses the lowest average bike count percentage at 7.89%, due to colder temperatures that may deter people from utilizing rental bikes as frequently. This seasonal breakdown highlights how biking patterns align with weather changes, contributing to the varying popularity of rental bikes throughout the year (see Figs. 9 and 11).

A violin plot is a visualization tool that effectively communicates the distribution of numerical data across different seasons [25]. In Fig. 14, the mirrored half-violin shape represents each season. The width of each violin corresponds to the density of data points at various temperature levels, creating a visual representation of the data's distribution. Thicker sections of the violin indicate a higher concentration of data points, while thinner sections signify lower concentrations [18]. The temperature records in the violin plot display the maximum, minimum, and average temperature ranges for each season. For autumn, the maximum temperature record stands at 30.5 °C, while the minimum reaches 3 °C, with an average of 14 °C. In spring, the maximum temperature is 29.4 °C, the minimum is −6.6 °C, and the average is 13 °C. Summer experiences higher temperature extremes, with a maximum of 39.4 °C, a minimum of 16.3 °C, and an average of 27 °C. Winter exhibits milder temperatures, with a maximum of 10.3 °C, a minimum of −17.8 °C, and an average of −3 °C.

5.8. Distribution and average of holidays vs bike count

In Seoul, the predominant utilization of rental bikes occurs primarily during non-holiday periods, where they serve as a popular alternative mode of transportation. This preference notably arises from the fact that people often seek weekends as periods of relaxation and weekdays are a time of professional commitments. The insights from

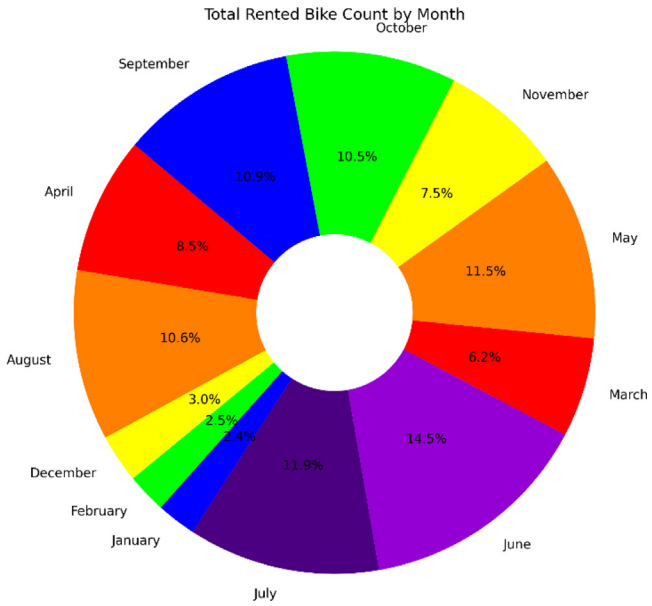


Fig. 12. Analysis of month vs total rented bike count.

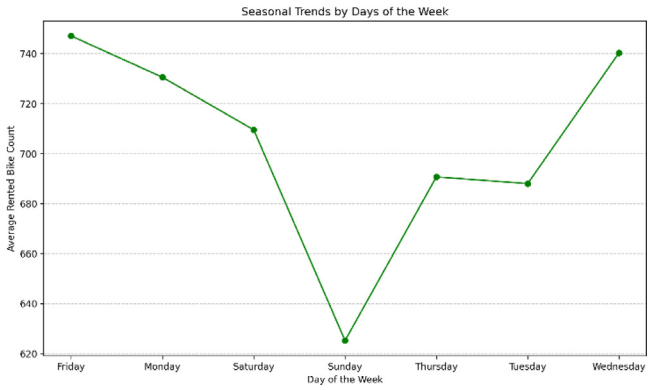


Fig. 13. Average rented bike count by days of the week.

Fig. 15 provide a crucial understanding of the dynamics of rental bike usage during different periods. It is evident that during holidays, the demand for rental bikes is lower, with a total count of 215,895 bikes rented and an average of about 500 (41.1%) bikes rented daily. In contrast, during non-holiday periods, the rental bike activity surges significantly, with a total of 5956,419 bikes rented and an average of approximately 715 (58.9%) bikes rented per day. These disparities highlight the importance of tailoring business strategies to accommodate customer preferences and needs during various times of the year (see Figs. 12 and 13)

5.9. Analysis of month vs total rented bike

The optimal seasons for visiting South Korea are the mild spring and autumn months. From April to June, the country becomes a pink-hued spectacle with cherry blossoms, azaleas, and lotuses in full bloom, particularly enhancing the national parks. It is important to note that this period can bring about higher crowds. It is advisable to avoid the summer months due to the uncomfortable humidity and the monsoon season. Autumn, spanning September to November, offers lower rainfall and more budget-friendly hotel rates. Despite the long and chilly winter, it holds its charm with a serene ambiance, the Hwacheon Sancheoneo Ice Festival, and opportunities for skiing. June stands out as the pinnacle month for tourism in South Korea,

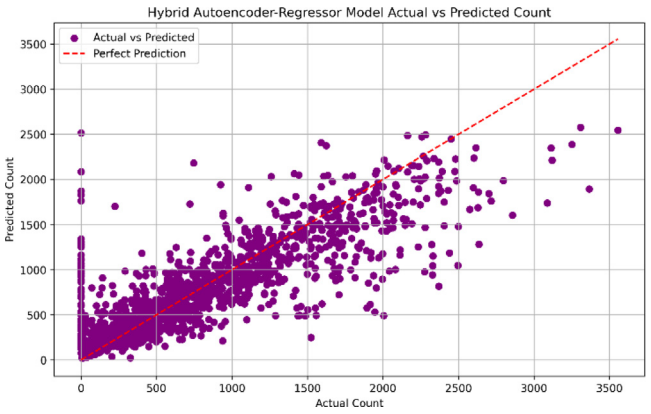


Fig. 14. Hybrid autoencoder regressor model.

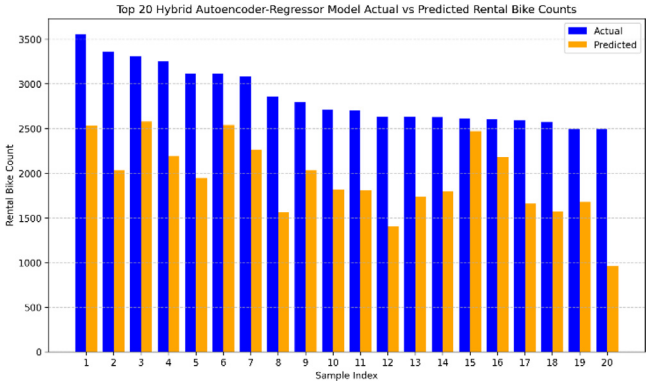


Fig. 15. Top 20 actual vs predicted rental bike count hybrid autoencoder regressor model.

particularly for international visitors [3]. This trend is reflected in the bike rental data, where June emerges as the peak period. Specifically, in June 2018, the bike rental count soared to 174,467 which means 11.4% of the total count, significantly contrasting to June 2017, which recorded only 0.4% which means 6,669 of total rentals. This variation underscores the substantial increase in tourist activity between these two years. Fig. 17 visually represents monthly rental bike counts for 2017 and 2018. The disparities become even more apparent. In 2018, the cumulative rental bike count reached 2240,904 which is 25.58% of the 2017 count while the count for 2017 was 87,600, based on the dataset. These figures elucidate the dynamic changes in tourist trends over the observed years. June captures the essence of heightened tourist engagement and exploration in South Korea. When examining the data annually, specific patterns emerge regarding bike rentals. The highest rental counts occurred in June, May, July, and October, each exceeding 600,000 rentals, and the lowest counts were observed in February, September, November, and December. These patterns provide valuable insights into the varying demand for bike rentals, driven by seasonal changes and the ebb and flow of tourism. By understanding these trends, bike rental businesses can tailor their services, promotions, and operational strategies to align with South Korea's distinct seasonal tourism dynamics (see Figs. 16 and 18)

5.10. Analysis of days vs total rented bike count

A notable observation is the relationship between the day of the week and the average rental bike count. On Sundays, the average count drops to 616, followed by 720 on Mondays, 678 on Tuesdays, 744 on Thursdays, 734 on Fridays, and 730 on Saturdays. This data signifies distinct fluctuations in rental bike demand based on different

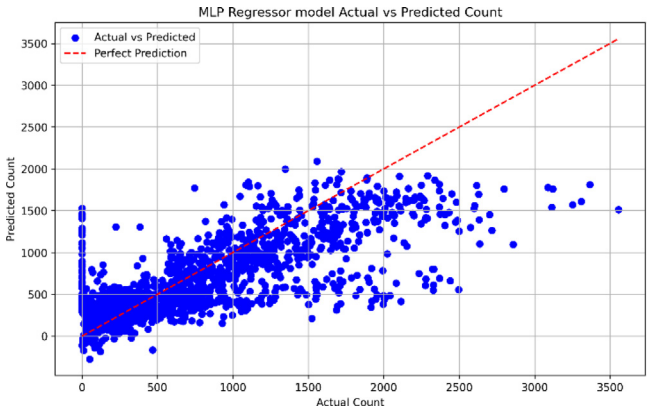


Fig. 16. MLP regressor model.

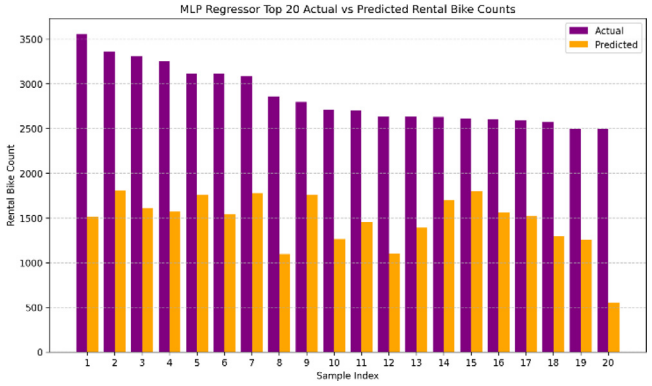


Fig. 17. Top 20 actual vs predicted rental bike count MLP regressor model.

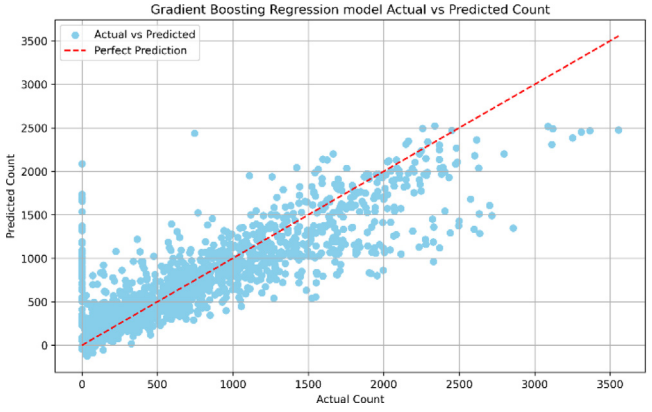


Fig. 18. Gradient boosting regression model.

days. Variations in weekend and weekday commute choices and leisure activities may be the reason for this trend. Interestingly, there is a slight decline in the rental bike count on Sundays, aligning with the common tendency for relaxation and reduced outdoor engagements during this day. This phenomenon underscores the significance of understanding human behavior and inclinations while crafting strategies for bike rental services. The dynamics of bike rental demand throughout the week have essential implications for the bike rental business. Unveiling rental patterns across time limits is pivotal for effective strategic planning and operational optimization. By closely analyzing daily, weekly, and monthly rental data, the business can identify peak rental periods, comprehend the influence of weather conditions and extraordinary events on customer behaviors, and make well-informed decisions

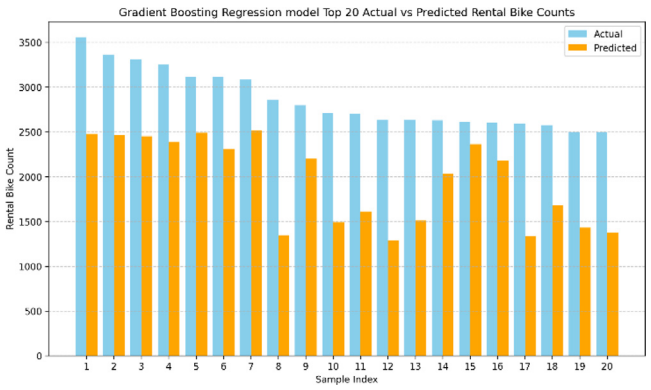


Fig. 19. Top 20 actual vs predicted rental bike count gradient boosting regression model.

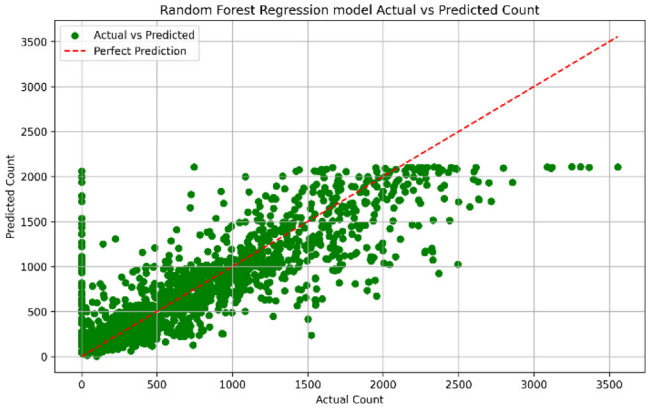


Fig. 20. Random forest regression model.

to enrich its services and enhance overall operational efficiency(see Figs. 19–22)

6. Metrics of prediction models

In regression analysis, R-squared (R^2) is a fundamental metric quantifying the proportion of variance in the dependent variable explained by independent variables. Higher R^2 values denote superior fit, indicating an effective explanation of variance. Conversely, lower Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) signify greater accuracy, suggesting closer prediction to actual values. A combination of these metrics provides a comprehensive evaluation of model performance, reflecting robustness and reliability. Notably, Random Search Cross-Validation (CV) emerges as a top performer, characterized by high R^2 and low MAE, MSE, and RMSE values, underlining its effectiveness in capturing complex data relationships.

6.1. Coefficient of determination (R^2 or R-squared)

In addition to the model-fit statistics, the R-square statistic is also commonly quoted and provides a measure that indicates the percentage of variation in the response variable that is 'explained' by the model. R-square, which is also known as the coefficient of multiple determination, is defined as the equation [19].

$$R^2 = 1 - \frac{\sum_{i=1}^{m} \left(X_i - Y_i\right)^2}{\sum_{i=1}^{m} \left(\bar{Y} - Y_i\right)^2} \tag{1}$$

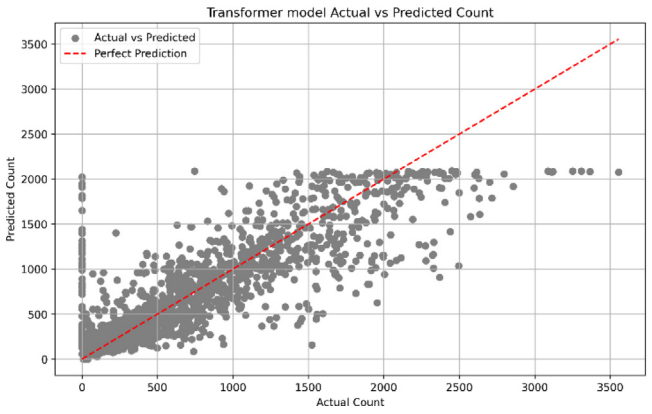


Fig. 21. Transformer model.

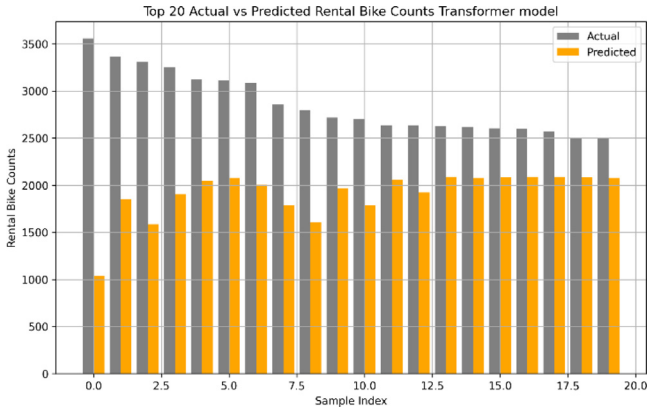


Fig. 22. Top 20 actual vs predicted rental bike count transformer model.

Where:

$$\bar{Y} = \frac{1}{m}\sum_{i=1}^{m} Y_i \tag{2}$$

m is the number of data points.
X_i is the predicted ith value.
Y_i is the actual ith value.

6.2. Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) is a metric used to measure the average magnitude of errors between predicted and actual values in a set of data. It is particularly useful in regression analysis. The formula for calculating Mean Absolute Error is:

$$MAE = \frac{1}{n}\sum_{i=0}^{n} |y_i - \hat{y}_i| \tag{3}$$

Where:
n is the number of data points.
y_i represents the actual value of i th data point.
$\hat{y}_i$ represents the predicted value of i th data point.
$\sum$ denotes summation across all the data points.

MAE gives a good sense of the average magnitude of errors in the rental bike demand predictions. It is measured in the same units as the original data, which makes it interpretable and easy to compare with the actual values. Lower MAE values indicate better predictive accuracy.

6.3. Mean Square Error (MSE)

The Mean Squared Error (MSE) is a widely used metric in statistics and machine learning to measure the average squared differences between predicted values and actual (observed) values. It is especially valuable for evaluating the accuracy of predictive models, such as regression models. The formula for MSE is:

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{n} \tag{4}$$

Where:
Σ is the summation symbol (summing over all data points).
y_i is the actual (observed) value for the i data point.
$\hat{y}_i$ represents the predicted value for the i data point.
n represents the total number of data points.

MSE is a way to quantify how well a predictive model performs by measuring how close its predictions are to the real data points. The metric encourages the model to minimize prediction errors, and it is particularly useful in regression tasks, where you aim to predict numerical values. Lower MSE values are generally desirable because they indicate a more accurate model [20].

6.4. Root Mean Squared Error (RMSE)

RMSE is a commonly used metric for assessing the accuracy of predictive models. It quantifies the average magnitude of errors between predicted values and actual values. RMSE is beneficial when you need to express prediction errors in the same units as the target variable, making it more interpretable. The formula for RMSE calculates the square root of the average of the squared differences between predicted and actual values. A lower RMSE indicates better accuracy in predicting target values. It is an essential tool in regression analysis and machine learning for evaluating model performance [20]. The formula for RMSE is:

$$RMSE = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n}} \tag{5}$$

Where:
RMSE: Root Mean Squared Error
y_i: Individual predicted values
ȳ: Mean of the actual values
n: Number of data points

6.5. Pearson correlation coefficient

The Pearson correlation coefficient, often denoted as Pearson's r or simply r, is a statistical measure of the linear relationship (correlation) between two continuous variables. It quantifies the strength and direction of the linear association between two variables, typically ranging from −1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no linear correlation [27]. The formula for calculating Pearson's correlation coefficient (r) between two variables X and Y is as follows:

$$r = \frac{\sum\left(\left(X - \overline{X}\right)\left(Y - \overline{Y}\right)\right)}{\left[\sqrt{\sum\left(X - \overline{X}\right)^2 \sum\left(Y - \overline{Y}\right)^2}\right]} \tag{6}$$

Where:
X and Y are data points.
$\overline{X}$ and $\overline{Y}$ are the means of X and Y, respectively.

7. Predictive modeling approaches

The primary objective of this project is to develop a robust predictive model for forecasting the Rented Bike Count in Seoul City. Initially,

exploration of the Ordinary Linear Regression model revealed that the dataset did not exhibit a linear relationship among its variables, making this approach less suitable for accurate predictions. Subsequently, the Random Forest Regression model was adopted to capture non-linear relationships within the data. This decision allowed for accounting for the intricate interactions between variables and improving the prediction accuracy of the rented bike count. Advancement to the Gradient Booster Regression model was pursued for better predictive performance. This sophisticated ensemble learning technique offers enhanced predictive capabilities and helps mitigate potential overfitting concerns. Leveraging the Gradient Booster model aimed to achieve superior results in predicting the rented bike count, contributing to more informed decision-making and improved business analytics for bike rental services in Seoul. The predictive software package utilized for these analyses employs the scikit-learn library within the Python programming environment to effectively implement the OLS regression, Random Forest, and Gradient Boosting, models. [28,29].

7.1. Hybrid autoencoder — regressor model

The Hybrid Autoencoder-Regressor Model merges an autoencoder and a regressor into a cohesive framework. Autoencoders, neural networks skilled at learning efficient data representations, employ an encoder to compress input into a lower-dimensional latent space and a decoder to reconstruct the original input. These models excel in dimensionality reduction, feature learning, and denoising tasks. Conversely, regressors, encompassing algorithms like linear regression and decision trees, predict continuous values based on input features, commonly used in forecasting house prices or sales volumes. The Hybrid Autoencoder-Regressor Model trains the autoencoder to encode and reconstruct input data while training the regressor to utilize the learned latent representations for predicting target variables. The autoencoder encodes new input into the latent space during inference, enabling the regressor to make predictions. Leveraging the autoencoder's feature learning enhances prediction accuracy, which is especially valuable for high-dimensional or noisy datasets where manual feature engineering is challenging. Joint optimization during training can outperform traditional regression models. Implementing an autoencoder-based regression model in TensorFlow involves building and training the autoencoder, extracting latent representations, training a regression model on these representations, and evaluating performance on a separate test dataset [30].

(1) Autoencoder Component — Encoder

$$Z = f(XW_1 + b_1) \tag{7}$$

Z – Encoded representation of input features.
X – Input features.
W_1 – Weight matrix of the encoder layer.
b_1 – Bias vector of the encoder layer.
f – Activation function (in this case, ReLU)

(2) Autoencoder Component — Decoder

$$\hat{X} = g\left(ZW_2 + b_2\right) \tag{8}$$

$\hat{X}$ – Reconstructed input features
Z – Encoded representation of input features.
W_2 – Weight matrix of the decoder layer.
b_2 – Bias vector of the decoder layer.
g – Activation function (in this case, ReLU).

(3) Loss Functions

$$\text{Autoencoder Loss } \mathcal{L}_{AE} = \left\|X - \hat{X}\right\|^2 \tag{9}$$

– Measures the reconstruction error between the original input features (X) and the reconstructed features ($\hat{X}$)

$$\text{Regressor Loss } \mathcal{L}_{Reg} = \|y - \hat{y}\|^2 \tag{10}$$

– Measures the prediction error between the true target variable (y) and the predicted target variable ($\hat{y}$) produced by the regressor.

$$\text{Hybrid Loss } \mathcal{L}_{Hybrid} = \alpha \cdot \mathcal{L}_{AE} + (1 - \alpha) \cdot \mathcal{L}_{Reg} \tag{11}$$

– α is a hyperparameter that balances the contribution of the autoencoder and regressor losses.

The performance assessment of the Hybrid Autoencoder-Regressor Model reveals promising results, indicating its effectiveness in capturing patterns within the dataset. The R-squared (R2) value of 0.709 denotes that approximately 70.9% of the variability in the target variable is accounted for by the model, showcasing its ability to explain a significant portion of the observed variance. A Mean Squared Error (MSE) of 121065.57 reflects the average squared disparity between predicted and actual values, with lower values indicating superior predictive performance. Meanwhile, the Root Mean Squared Error (RMSE) of 347.94, derived from the square root of the MSE, provides insight into the typical magnitude of prediction errors, where smaller values signify better model fit. Furthermore, the Mean Absolute Error (MAE) of 213.86, representing the average absolute deviation between predicted and actual values, offers a more interpretable measure of model accuracy. Additionally, a Pearson Correlation Coefficient of 0.843 indicates a strong positive linear association between predicted and actual values, reinforcing the model's capacity to capture underlying data patterns effectively. In summary, these metrics collectively affirm the Hybrid Autoencoder-Regressor Model's proficiency in predicting the target variable, underscored by its commendable accuracy and explanatory capability.

7.2. Multi-Layer Perceptron (MLP) Regressor

The Multi-Layer Perceptron (MLP) Regressor is a feedforward artificial neural network utilized for regression tasks in predictive analysis. It encompasses multiple layers of interconnected nodes, comprising an input layer, one or more hidden layers, and an output layer. Each neuron in the network computes a weighted sum of its inputs, followed by applying an activation function to generate the output. The core concept of the MLP Regressor involves several key steps. Firstly, the Input Layer comprises neurons that receive the feature values of the input data, with each neuron corresponding to a feature. The number of neurons equals the dimensionality of the input data. Secondly, the Hidden Layers are intermediate layers situated between the input and output layers, responsible for capturing intricate patterns and relationships within the data. Neurons in these layers receive inputs from all neurons in the previous layer and produce an output based on a weighted sum of these inputs. Thirdly, the Output Layer contains one or more neurons that generate the model's predictions. In regression tasks, there is usually one neuron, and its output represents the predicted continuous value. Fourthly, at each neuron, inputs are multiplied by corresponding weights, and the weighted sum of these inputs is computed. Additionally, a bias term may be added to this sum. Fifthly, the weighted sum undergoes an activation function, introducing non-linearity to capture complex relationships in the data. Common activation functions include Rectified Linear Unit (ReLU), Hyperbolic Tangent (tanh), and Sigmoid. Sixthly, during the Training phase, the model learns optimal parameter values (weights and biases) by minimizing a predefined loss function, such as Mean Squared Error (MSE) or Mean Absolute Error (MAE). This process typically involves optimization algorithms like Gradient Descent. Lastly, once trained, the MLP Regressor predicts new, unseen data by passing inputs through the network and obtaining outputs from the output layer [31,32].

The performance evaluation of the Multi-Layer Perceptron (MLP) Regressor model indicates a satisfactory fit to the dataset. With an R-squared (R2) value of 0.579, the model accounts for approximately 57.9% of the variance in the target variable, implying its ability to capture a significant portion of the inherent data patterns. The Mean Squared Error (MSE) of 175468.49 quantifies the average squared

deviation between predicted and actual values, where lower values suggest better model performance. The Root Mean Squared Error (RMSE) of 418.89, derived from the square root of the MSE, provides an average magnitude of errors, with lower values indicating superior fit. The Mean Absolute Error (MAE) of 275.07 signifies the average absolute difference between predicted and actual values, offering a more understandable measure of model accuracy. Moreover, the Pearson Correlation Coefficient of 0.778 highlights a robust positive linear association between predicted and actual values, further reinforcing the model's proficiency in capturing inherent data trends. These metrics collectively indicate that the MLP Regressor model performs adequately in predicting the target variable, demonstrating moderate accuracy and explanatory capability.

$$z = \sum_{i=1}^{n} \omega_i \cdot x_i + b \tag{12}$$

Where:
z is the weighted sum,
ω_i are the weights associated with each input x_i,
b is the bias term,
n is the number of inputs.

Next, the weighted sum is passed through an activation function f to introduce non-linearity into the model. Common activation functions include the Rectified Linear Unit (ReLU), Hyperbolic Tangent (tanh), and Sigmoid functions. The choice of activation function depends on the specific problem and network architecture.

$$a = f(z) \tag{13}$$

Where:
a is the output of the activation function.

The network output is computed based on the activations of the neurons in the output layer. There is typically one neuron in the output layer for regression tasks, and its output represents the predicted continuous value.

$$\hat{y} = g(a) \tag{14}$$

Where:
$\hat{y}$ is the predicted output,
g is an optional transformation function applied to the output of the last layer.

7.3. Gradient boosting regression model

GBM is an ensemble or iterative learning model that uses decision trees for regression. GBM uses a technique called gradient-boosted trees to create a model in which every predictor corrects the predecessor's error; as this loss gradient is minimized, the model is fit. The GBM algorithm is expressed as an additive model of decision trees in the equation.

$$f_m(x) = \sum_{m=1}^{M} T\left(x, \theta_m\right), T\left(x, \theta_m\right) \tag{15}$$

where θ_m is the parameter of the decision tree and M is the number of decision trees. The algorithm then uses a loss function to optimize the next learner parameter shown by the equation.

$$\theta_{m+1} = argmin \sum_{m=1}^{N} L\left(y_t, F_m(x) + h\left(x, \theta_{m+1}\right)\right) \tag{16}$$

Meticulous training of each successive model is conducted to minimize the loss function of its predecessor through gradient descent, which involves metrics like mean squared error or cross-entropy in the following equation. In the Seoul rented bike dataset context, the interplay among variables does not adhere to a linear relationship, substantiated by the modest R-squared value of 0.579 that emerged from the MLP Regressor model. Given the linear assumptions inherent to MLPR, its applicability falters in grasping the intricate and non-linear intricacies ingrained within the data structure. Hence, the recourse gravitated toward adopting the Gradient Boosting Regression model, which emerges as a more fitting alternative in this context [33]. Here, the R-squared value (0.747) explains the proportion of the variance in the dependent variable (rented bike count) that the independent variables (features) in the model account for. A higher R-squared value indicates a better fit of the model to the data, meaning that the independent variables can explain approximately 74.7% of the rented bike count variation.

In the analysis of the Gradient Booster Regression model, several key evaluation metrics were considered. The R-squared (R2) value of 0.747 suggests that approximately 74.7% of the variability in the rented bike count can be explained by the model's features. The Mean Squared Error (MSE) of 105340.97 and Root Mean Squared Error (RMSE) of 324.56 indicate the average squared and square root of the differences between predicted and actual values, respectively. Lower values of MSE and RMSE imply better prediction accuracy. Additionally, the Mean Absolute Error (MAE) of 205.31 signifies the average absolute difference between predicted and actual values. A lower MAE is desirable for improved model performance. Finally, the Pearson Correlation Coefficient of 0.866 illustrates a strong positive linear relationship between predicted and actual values. This comprehensive evaluation provides valuable insights into the model's effectiveness in predicting rented bike counts and guiding potential enhancements or adjustments to optimize predictive accuracy [34].

The decision to transition to another model, such as Random Forest Regression, arose from the aspiration to enhance the accuracy of predicting the rented bike count. Random Forest Regression is known for handling non-linear relationships between variables, which might be the case in the dataset. Random Forest Regression is expected to capture more complex patterns and interactions between the features, potentially leading to better predictions and reducing the Mean Squared Error.

7.4. Random forest regression model

By leveraging ensemble learning and combining multiple decision trees, Random Forest can manage non-linear relationships in the data. The R-squared value achieved by the Random Forest model is significantly higher, indicating a better fit to the data and improved predictive performance. Capturing intricate interactions between the features with Random Forest Regression leads to more accurate predictions of the rented bike count in Seoul. The obtained R-squared value of 0.7703 indicates that approximately 77.03% of the variability in the rented bike count is explained by the features used in the model. A higher R-squared value is desirable as it signifies a better fit of the model to the data. The Mean Squared Error (MSE) of 95,675 and Root Mean Squared Error (RMSE) of 309.31 indicate the average squared and square root of the differences between predicted and actual values, respectively. The Mean Squared Error (MAE) value of 184.17 represents the average squared difference between the actual and predicted bike counts. A lower MSE indicates that the model's predictions are close to the actual values, which is suitable for accurate predictions. The Pearson Correlation Coefficient of 0.8784 indicates a strong positive linear relationship between the predicted and actual bike counts. This suggests that the model's predictions are closely related to the true bike counts, indicating reliable model performance. The Random Forest Regression model demonstrates strong predictive capabilities for forecasting the rented bike count in Seoul city, supported by its high R-squared value, low MSE, and strong Pearson Correlation Coefficient [35].

7.5. Regression Transformer model

The Regression Transformer (RT) model introduced in the provided text represents a fresh approach to regression challenges by refraining

Table 1
Machine learning model comparisons.

Model	R2	MSE	RMSE	MAE	Pearson CC
Hybrid Autoencoder-Regressor Model	0.709	121065.57	347.94	213.86	0.843
Multi-Layer Perceptron Regressor	0.579	175468.49	418.89	275.07	0.778
Gradient Booster Regression	0.747	105340.97	324.56	205.31	0.866
Random Forest Regression	0.77	95 675	309.31	184.16	0.878
Transformer Model	0.731	111938.4	334.57	209.3	0.855

from directly modeling target variables as functions of input variables. Instead, it reimagines regression as a sequence modeling task focusing on textual tokens, effectively blurring the distinction between predictive and conditional generative models. The RT, as a class of models, is adept at processing both numerical and textual tokens, allowing it to seamlessly handle property prediction and conditional text generation. Unlike conventional methods that rely on dense layers and regression loss, the RT utilizes tokenization of numbers and cross-entropy loss to achieve successful regression outcomes. Through a training process that alternates between optimizing property prediction and text generation, the RT is imbued with an inductive bias tailored for managing continuous properties. Noteworthy is the RT's capability to generate text sequences conditioned on continuous properties, underscoring its adaptability. With its goal to narrow the semantic gap between property prediction and generative models in fields such as chemo- and bioinformatics, the RT provides a unified framework catering to both regression tasks and conditional text generation [36].

8. Machine learning model comparisons

In this context, three distinct machine learning models were employed: Hybrid Autoencoder-Regressor Model, Multi-Layer Perceptron Regressor, Gradient Booster Regression, Random Forest Regression, and Transformer Model. Table 1 presents the comparative analysis of these models, assessing metrics such as *R*-square values, Mean squared error values, and correlation coefficient values. This approach is driven by the absence of correlation among variables within the Seoul rental bike dataset, guiding toward a comprehensive evaluation of the models' performance. The evaluation phase of a machine learning model assumes particular significance during these stages, furnishing crucial insights into the predictive accuracy of the models. Precision metrics serve as a crucial benchmark for assessing the effectiveness and viability of deploying the machine learning model (see Figs. 24–26)

Fig. 23 represents the *R*-squared value measures of how well a regression model fits data. A higher *R*-squared means a better fit and lower Mean Squared Error (MSE). It ranges from 0 to 1, with one indicating a perfect fit. Pearson's correlation coefficient (r) gauges linear relationships between variables. It varies from −1 to +1, where −1 and 1 represent perfect linear relationships. Stronger correlations have higher absolute values. A coefficient of 0 means no linear relationship. The sign indicates direction: positive values show variables rise together, and negative values show one falls as the other rises.

Transitioning to MAE, Gradient Boosting, Random Forest model, and Transformer Models demonstrate lower values than other models, with Random Forest showcasing the smallest MAE. This implies that Random Forest yields more accurate predictions with smaller discrepancies between predicted and actual values. Furthermore, examining the Pearson Correlation coefficient, Gradient Boosting, Random Forest

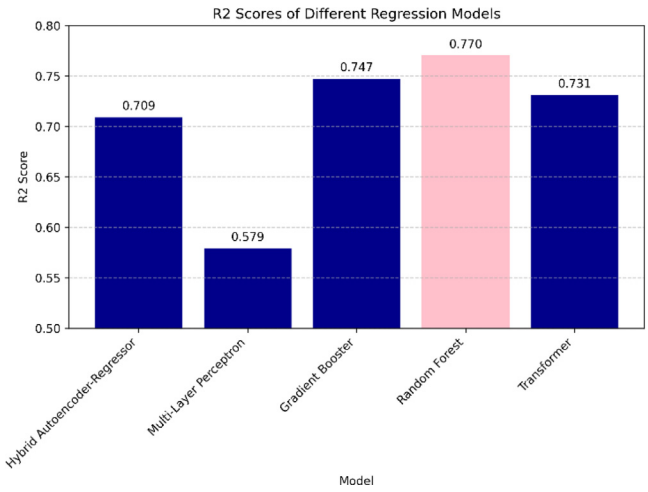


Fig. 23. Comparison of *R*-squared values.

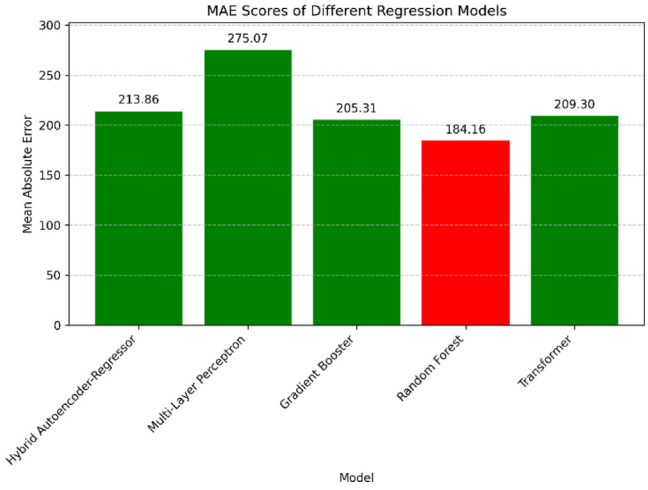


Fig. 24. Comparison of mean absolute error values.

model, and Transformer Models present notably higher coefficients compared to other models, suggesting stronger linear relationships between predictors and the target variable. Remarkably, Random Forest displays the highest Pearson Correlation coefficient, indicating the most robust correlation between variables. In summary, Gradient Boosting, Random Forest, and Transformer models surpass other models across all metrics, with Random Forest showing the highest overall performance [37].

9. Conclusion

The analysis aimed to predict the number of rented bikes in Seoul city based on weather and time-related features. The analysis began with a thorough exploration of the dataset, examining its distribution, and establishing relationships between variables. Key features include temperature, humidity, wind speed, visibility, solar radiation, rainfall, snowfall, hours of the day, seasons, holidays, and functioning days. It was observed that while certain variables exhibited linear relationships, others did not conform to such patterns. Among the models evaluated, the performance metrics indicate that the Random Forest Regression model stands out as the most effective for predictive analysis. With an *R*-squared value of 0.77, it explains a substantial portion of the variance

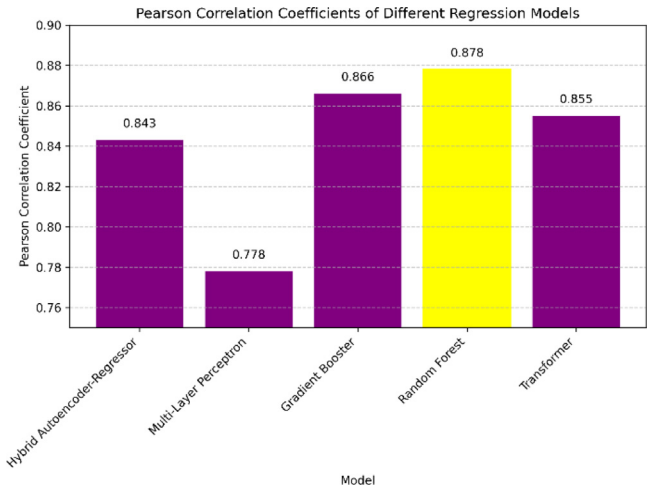


Fig. 25. Comparison of Pearson correlation coefficient values.

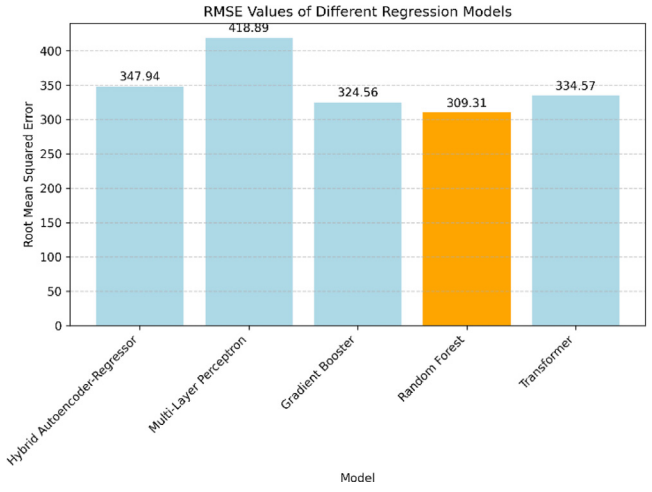


Fig. 26. Comparison of RMSE values.

in the target variable. The model also exhibits the lowest Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), indicating superior accuracy in predicting the target variable compared to other models. Furthermore, the Pearson Correlation Coefficient, which measures the linear relationship between predicted and actual values, is highest for the Random Forest Regression model, indicating a strong correlation. Therefore, based on these metrics, it can be concluded that the Random Forest Regression model is the most suitable choice for accurate predictive analysis in this context. The comparative analysis establishes the Random Forest Regression model as the most suitable and accurate for predicting rented bike counts in Seoul. Its superior performance and adeptness in managing non-linear relationships make it an optimal choice for this dataset. In conclusion, this study underscores the critical importance of selecting appropriate regression models to ensure accurate predictions. The insights derived can empower bike rental businesses in Seoul to optimize resources, anticipate demand fluctuations, and enhance operational efficiency. Future endeavors will focus on implementing additional algorithms to further enhance predictive accuracy and integrate prescriptive analysis to offer actionable recommendations for rental bike businesses, thereby elevating operational efficiency and service quality. The commitment to advancing research in this field remains steadfast, with a vision to drive improvements benefiting both the industry and the community it serves.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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